Muhammad Talha Iqbal

(Software Engineer · AI Security · Researcher · Entrepreneur)

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EDUCATION

Bachelor of Science in Software Engineering Jan 2023 – Present

COMSATS University Lahore, Lahore, Pakistan

- Relevant Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Information Security, Pattern Recognition, Human–Computer Interaction, Parallel & Distributed Computing, Software Project Management.

Cambridge A-Levels and O-Levels Completed 2023

International School Lahore, Lahore, Pakistan

- Merit Scholarship recipient.

EXPERIENCE

Founding Partner & Operations Lead 2024 – Jun 2026

***Cirkles** — Event Discovery Platform, Lahore, Pakistan*

- Joined as a founding partner with operational ownership across launch planning, partner outreach, and team coordination during pre-launch and early growth phases.
- Built and maintained feedback loops with event organisers and early users; translated qualitative input into operational priorities for the product and engineering teams.
- Coordinated cross-team workflows between design, engineering, and research collaborators to maintain release tempo through high-pressure build cycles.
- Owned launch operations, partnership pipelines, and user-acquisition experiments; held the operating function end-to-end across the venture's lifecycle.
- Drove organisational decisions through structured analysis of team alignment, market signal, and product trajectory, including the decision to exit in mid-2026.

Data Science Intern Jun 2025 – Aug 2025

***Systems Limited** — AI & Analytics Practice, Lahore, Pakistan*

- Joined the AI & Analytics practice on a focused data-science engagement; worked on predictive analytics and ML model development for internal product use-cases, with a secondary contribution to UX research for analytical dashboards.
- Built end-to-end supervised-learning pipelines (scikit-learn, XGBoost, LightGBM) for tabular operational data; covered EDA, missing-value imputation, feature engineering, model selection via stratified k-fold cross-validation, and threshold tuning on imbalanced targets.
- Developed time-series forecasting baselines (ARIMA, Prophet, gradient-boosted regressors with lag features) and benchmarked them on internal demand-signal datasets; produced reproducible Jupyter notebooks and model cards consumed by senior data scientists.
- Worked alongside the design pod on UX flows for an analyst-facing dashboard; produced Figma wireframes and supported a small round of qualitative user-research interviews with internal analyst personas.
- **Key learnings:** end-to-end ML lifecycle in an enterprise setting; rigour around data leakage, validation strategy, and metric selection (ROC-AUC, PR-AUC, MAPE); exposure to MLOps fundamentals — experiment tracking, model versioning, and basic monitoring patterns; how production analytics teams translate model output into product decisions.

Sales & Business Development Lead Jan 2023 – Present

***U.n.I Global** — ED Tech Startup, Lahore, Pakistan*

- Led sales and business development for a university-focused connectivity platform that scaled to 8,000+ active users within six months of launch.
- Built and executed campus-level outreach strategy across multiple Pakistani universities; conducted on-ground user research to surface validated student pain points and inform product positioning.
- Established partnership pipelines with university societies and student bodies, driving community-led acquisition without paid marketing spend.
- Collaborated with the product team on onboarding flows, retention experiments, and engagement loops grounded in user-research findings.
- Supervised junior team members across outreach, campus operations, and content functions; ran weekly performance reviews tied to acquisition and engagement metrics.

Data Analyst Intern Jul 2024 – Sep 2024
Systems Limited and PartnerLinq *(US-facing Supply-Chain Product), Lahore, Pakistan*

- Embedded within the PartnerLinq product team — a US-facing supply-chain integration platform — on workforce analytics and resource-allocation systems for the Operations function.
- **Independently designed and built a Human Resource Management System (HRMS)** for the Operations group: defined the relational schema in SQL Server, implemented CRUD workflows, and integrated the system with the existing reporting pipeline for end-to-end allocation visibility.
- Engineered Power BI dashboards (DAX, SQL) tracking onboarding pipelines, team utilisation, and capacity allocation across delivery pods; authored optimised DAX measures for incremental refresh on large fact tables.
- Exposed to workflow-orchestration concepts including DAG-based pipeline structuring and automation patterns used in operational data flows; mapped these against the platform's existing ETL stack.
- Translated raw operational data into structured reporting consumed directly by project coordinators across multiple agile pods.
- **Key learnings:** working inside a US-facing enterprise product environment; relational data modelling discipline; the bridge between ad-hoc analytics and shipped internal tooling; the difference between dashboards built for analysts and dashboards built for operational decision-makers.

Founder & 3D Design Lead Jul 2019 – 2023
Pumpkin Studios *— Independent 3D Visualisation Studio, Lahore, Pakistan*

- Founded an independent 3D visualisation studio at age 12; operated through secondary school across enterprise and government client engagements.
- Delivered photorealistic product renders and architectural visualisations for clients including Toyota Indus Motors, Dawn Bread, and the Pakistan Army.
- Managed end-to-end project pipelines independently — client briefing, scoping, production, revisions, and delivery — across concurrent engagements under strict branding and deadline constraints.

ACADEMIC PROJECTS AND RESEARCH

Mitigation of Privilege Escalation in Autonomous AI Agents Sep 2025 – Present
Research and Final Year Project *— COMSATS University Lahore, Lahore, Pakistan*

- Designing a multi-layer verification framework — the *Federated Agentic Protocol (FAP)* — targeting prompt-injection-driven privilege escalation and the **confused-deputy problem** in LLM-based tool-calling environments, where an authenticated agent is coerced by adversarial inputs into invoking privileged tools on behalf of a lower-trust principal.
- Architected an isolated **verifier-LLM layer** that sits between the planning agent and the tool-execution runtime: the verifier never sees raw user input or untrusted tool output, but consumes only *structured metadata* (root intent, current tool argument set, prior action-chain summary, declared scope) and returns a constrained-JSON alignment verdict — breaking the implicit-authority chain that prompt-injection exploits depend on.
- Implemented the verifier using **Phi-3 Mini (3.8B)** hosted locally via Ollama with temperature pinned at 0.1 and JSON-schema-constrained decoding for deterministic, auditable verdicts; benchmarked against a stronger 7B baseline to validate that a small, isolated verifier is sufficient under metadata-only inputs.
- Built a **YAML-based declarative policy engine** validated through Pydantic: enforces tool whitelists, parameter-level validators, scope constraints mapped to root-intent categories, and session-scoped rate limits at sub-5 ms decision latency on the hot path.
- Implemented a **cryptographic hash-chain audit log** providing tamper-evident accountability across all agent actions, enabling post-hoc verification of agent decision integrity and incident reconstruction.
- Evaluation against established agentic-security benchmarks (AgentDojo-style injection suites and bespoke confused-deputy scenarios) is in progress; framework packaged for open-source release as a Python SDK with a one-line developer integration via the `@IntentGate` decorator pattern.

Cryptanalysis and Design of Substitution–Permutation Block Ciphers May 2025
Information Security Project *— COMSATS University Lahore, Lahore, Pakistan*

- Designed, implemented, and cryptanalysed a custom **substitution–permutation network (SPN)** block cipher (64-bit block, 80-bit key, 8 rounds) modelled on PRESENT and a simplified AES; full reference implementation written in Python with a Rust comparison build for performance benchmarking.
- Constructed two candidate **4×4 S-boxes** and evaluated each against the standard cryptographic criteria: *nonlinearity*, *differential uniformity*, *algebraic degree*, and *strict avalanche criterion*; selected the candidate with differential uniformity 4 and nonlinearity 4 and documented the trade-offs against the AES 8×8 S-box.
- Performed **differential cryptanalysis** on reduced-round (3- and 4-round) variants: enumerated input differences, built the difference distribution table (DDT) for the chosen S-box, propagated differential characteristics through the linear layer, and recovered partial last-round key bits using a chosen-plaintext attack with measured data complexity.

- Performed **linear cryptanalysis** in parallel: built linear approximation tables (LAT), traced biased linear trails through the cipher, and reproduced Matsui's Algorithm-2 key-recovery on reduced rounds; cross-validated empirical bias against theoretical predictions.
- Benchmarked the resulting cipher against **AES-128** and **PRESENT-80** on throughput, key-schedule cost, and resistance to the same attack suite; produced a written report and a reproducible Jupyter notebook with all DDT/LAT computations, attack scripts, and SageMath verification of S-box properties.

InsightHub — Brand–Consumer Review & Structured Discussion Platform May 2024
Academic Project *— COMSATS University Lahore, Lahore, Pakistan*

- Designed and built **InsightHub**, a two-sided web platform that connects consumer brands with their end users for *structured product reviews and moderated discussion threads*, with the explicit goal of giving brand product-teams a higher-signal alternative to scattered social-media feedback.
- Implemented the system as a full-stack web application: **React.js** frontend, **Node.js / Express** REST API, **MongoDB** primary store, and **Firebase Auth** for identity — with role-based access control separating consumer accounts, brand-side reviewer accounts, and brand admin dashboards.
- Built a **verified-reviewer flow** (lightweight purchase-proof attestation) and a moderated discussion-thread model with threaded replies, voting, and tag-based topic clustering; reviews are first-class entities with structured fields (sentiment, feature mentioned, severity) on top of free-form text.
- Integrated a **lightweight NLP pipeline** (sentiment classification + keyphrase extraction using a HuggingFace transformer served via a Python micro-service) that auto-tags incoming reviews and feeds an aggregated insight view consumed by the brand-side dashboard.
- Developed a **brand analytics dashboard** surfacing review velocity, sentiment trends, top recurring complaints, and feature-level satisfaction — designed iteratively against three early brand-side users sourced from the U.n.I network.
- Conducted a structured user-research study with ∼25 participants (consumers and brand operators) covering onboarding friction, trust signals, and willingness to leave structured vs. unstructured reviews; findings drove a redesign of the review-submission flow.
- Documented system design, threat model (review fraud, brand-side moderation abuse, scraping), and data-model decisions; project earned distinction-level marks in the Human–Computer Interaction and Database Systems course strands.

CERTIFICATIONS

- **Machine Learning Specialization** — Stanford University & DeepLearning.AI (Coursera)
- **Ethical Hacker intro** — Cisco Networking Academy
- **Cybersecurity Learning Path** — TryHackMe
- **UX Design fundamentals** — Google (Coursera)

TECHNICAL SKILLS

Languages	Python, Dart, Java, JavaScript, C++, SQL
Web & Mobile	Flutter, React.js, Node.js, Express, HTML, CSS
Databases & Cloud	MongoDB, Firebase, AWS, SQL Server
AI & Security	Local LLM deployment (Ollama), agentic-system architecture, prompt-injection defence, LangChain, verifier-LLM design, ethical hacking, penetration-testing fundamentals, applied cryptography (SPN ciphers, differential & linear cryptanalysis)
Data & ML	scikit-learn, XGBoost, LightGBM, pandas, NumPy, Jupyter, time-series forecasting, feature engineering, model evaluation
Data & Analytics	Power BI (DAX), data visualisation, AI-assisted dashboards, operational analytics
Design	UI/UX design, Figma, wireframing, user research, 3D modelling and visualisation
Tools & Frameworks	Git/GitHub, REST APIs, Pydantic, FastAPI, VS Code, workflow orchestration (DAG-based pipelines)
Management	Team leadership, project management, event management, Agile/Scrum, user research, stakeholder coordination

HONOURS & DISTINCTIONS

- **Silver Medal**, Microsoft Hackathon — hosted by RAS Society
- **Winner**, LUMS Psifi Science & Technology Festival — 2022, 2023
- **3rd Place**, COLABS Pitch to Win — 2025